



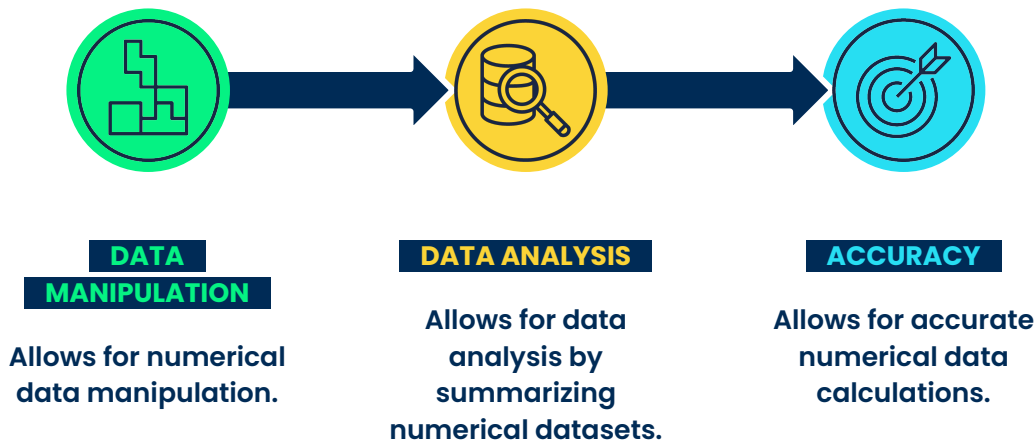
SQL numeric functions

# Numeric functions in SQL

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# The importance of numeric functions in SQL

Numeric functions are **built-in functions that operate on numeric data types** (such as integers, decimals, and floating points) and perform various **mathematical and statistical operations** on them.



# Introduction to numeric functions in SQL

## Aggregate numeric functions

- `MIN()`
- `MAX()`
- `AVG()`
- `SUM()`
- `COUNT()`

## Scalar numeric functions

- `ROUND()`
- `SQRT()`
- `LOG()`



Scalar numeric functions are not limited to the above. These will be discussed in this section.

# General Household Survey (GHS) dataset

| Province      | Y_2018    | Y_2019    | Y_2020    | Y_2021    | Y_2022    |
|---------------|-----------|-----------|-----------|-----------|-----------|
| Eastern_cape  | 23612.117 | 24218.898 | 25785.763 | 27074.594 | 28049.845 |
| Free_state    | 25974.478 | 27897.917 | 29459.302 | 30932.342 | 48532.912 |
| Gauteng       | 31748.937 | 33112.344 | 34966.168 | 36715.740 | 38378.338 |
| KwaZulu-Natal | 26188.484 | 27496.383 | 29141.264 | 30598.355 | 31914.132 |
| Limpopo       | 30140.911 | 31726.672 | 33502.167 | 35178.147 | 36820.465 |
| Mpumalanga    | 27321.994 | 28283.472 | 29867.734 | 31360.122 | 32963.496 |
| Northwest     | 25522.502 | 27621.283 | 29168.121 | 30626.714 | 32500.753 |
| Northern_cape | 27833.420 | 29288.380 | 30929.224 | 32475.897 | 33222.342 |
| Western_cape  | 30164.275 | 31863.417 | 33647.873 | 35329.837 | 36919.034 |

This dataset shows the **average monthly salaries of public servants** obtained from 2018 to 2022 by province.

We will use it to illustrate **different numeric functions**.

Name: Salaries.ghs\_db

# The MIN() function

The MIN() function returns the **smallest or lowest value** of the selected column. Suppose we want to calculate the **lowest average monthly salary** made in the year 2020.

Query

```
SELECT  
    MIN(Y_2020) AS Lowest_salary_2020  
FROM  
    Salaries.ghs_db;
```

Output

| Lowest_salary_2020 |
|--------------------|
| 25789.763          |

| Y_2020    |
|-----------|
| 25785.763 |
| 29459.302 |
| 34966.168 |
| 29141.264 |
| 33502.167 |
| 29867.734 |
| 29168.121 |
| 30929.224 |
| 33647.873 |

# The MAX() function

The MAX() function returns the **largest or highest value** of the selected column. Suppose we want to calculate the **highest average monthly salary** made in the same year, 2020.

Query

```
SELECT  
    MAX(Y_2020) AS Highest_salary_2020  
FROM  
    Salaries.ghs_db;
```

Output

Highest\_salary\_2020

34966.168

| Y_2020    |
|-----------|
| 25785.763 |
| 29459.302 |
| 34966.168 |
| 29141.264 |
| 33502.167 |
| 29867.734 |
| 29168.121 |
| 30929.224 |
| 33647.873 |

# The AVG() function

The AVG() function returns the **average value** of the selected numeric column. Suppose we want to calculate the **average monthly salary** made in the year 2022.

Query

```
SELECT
    AVG(Y_2022) AS Average_salary_2022
FROM
    Salaries.ghs_db;
```

Output

Average\_salary\_2022

35477.92419

Y\_2022

28049.845

48532.912

38378.338

31914.132

36820.465

32963.496

32500.753

33222.342

36919.034

# The SUM() function

The SUM() function returns the **total sum** of a specified numeric column. Suppose we want to find the sum of monthly salaries made in the year 2019.

Query

```
SELECT
    SUM(Y_2019) AS Total_salaries_2022
FROM
    Salaries.ghs_db;
```

Output

Total\_salaries\_2019

319301.3177

Y\_2019

24218.898

27897.917

33112.344

27496.383

31726.672

28283.472

27621.283

29288.380

31863.417



# The COUNT() function

The COUNT() function returns the **number of rows** of a specified column. Suppose we want to know how many provinces were used in this dataset.

## Query

```
SELECT
    COUNT(Province) AS Number_of_provinces
FROM
    Salaries.ghs_db;
```

## Output

| Number_of_provinces |
|---------------------|
| 9                   |

| Province      |
|---------------|
| eastern_cape  |
| free_state    |
| gauteng       |
| kwazulu_natal |
| limpopo       |
| mpumalanga    |
| northwest     |
| northern_cape |
| western_cape  |

# The COUNT(DISTINCT column) function

The COUNT(DISTINCT column) function returns the distinct or unique **number of rows** of a specified column. Suppose we want to know how many unique provinces were used in this dataset.

## Query

```
SELECT  
    COUNT(DISTINCT Province) AS Number_of_provinces  
FROM  
    Salaries.ghs_db;
```

## Output

| Number_of_provinces |
|---------------------|
| 9                   |

The unique number of provinces is equal to the number we got from the previous example. This is because there are no duplicate provinces in the Provinces column.

# The ROUND() function

The ROUND() function **rounds a numerical value to a specified number of decimal places**.  
The syntax is as follows:

```
SELECT ROUND( numerical value , decimal places )
```

This clause specifies the numerical value to be rounded.

This clause refers to the number of decimal places to round the numerical value to.



# The ROUND() function

Suppose we want to round the **average monthly salary** made in the year 2022 we calculated before to the nearest cent.

## Query

```
SELECT
    ROUND(Y_2022,2) AS Rounded_salaries
FROM
    Salaries.ghs_db
LIMIT 3;
```

## Output

| Rounded_salaries |
|------------------|
| 23612.12         |
| 25974.48         |
| 31748.94         |

| Y_2022   | Y_2022    |
|----------|-----------|
| 23612.12 | 23612.117 |
| 25974.48 | 25974.478 |
| 31748.94 | 31748.937 |
| 26188.48 | 26188.484 |
| 30140.91 | 30140.911 |
| 27321.99 | 27321.994 |
| 25522.50 | 25522.502 |
| 27833.42 | 27833.420 |
| 30164.28 | 30164.275 |

# The SQRT() function

The SQRT() function returns **the square root of a numerical value**. The syntax is as follows:

```
SELECT SQRT( numerical value )
```

This clause specifies the numerical value to be square rooted.

# The SQRT() function

The SQRT() function returns **the square root of a numerical value**. The syntax is as follows:

## Query

```
SELECT
    SQRT(Y_2021) AS Square_root_y_2021
FROM
    Salaries.ghs_db
LIMIT 3;
```

## Output

| Square_root_y_2021 |
|--------------------|
| 160.5794607...     |
| 171.6371221...     |
| 186.9924267...     |

| Y_2021    | Y_2021    |
|-----------|-----------|
| 160.58... | 27074.594 |
| 171.64... | 30932.342 |
| 186.99... | 36715.740 |
| 170.71... | 30598.355 |
| 183.04... | 35178.147 |
| 172.82... | 31360.122 |
| 170.79... | 30626.714 |
| 175.87... | 32475.897 |
| 183.43... | 35329.837 |

# The LOG() function

The LOG(base, numerical) function returns the **logarithm of a numeric value with a specified base**. The syntax is as follows:

```
SELECT LOG( numerical_value, base );
```

This clause specifies the numerical value to be logged.

This clause specifies the base of the logarithm of the numerical value.

# The LOG() function

Suppose we want to find the `log` of the average salaries and then round it to 2 decimal places.

## Query

```
SELECT
    LOG(Y_2021,10) AS Log_y_2021
FROM
    salaries.ghs_db
LIMIT 3;
```

## Output

| Log_y_2021     |
|----------------|
| 4.564852289... |
| 4.485698078... |
| 4.546272965... |

| Y_2021      | Y_2021    |
|-------------|-----------|
| 4.564852289 | 27074.594 |
| 4.485698078 | 30932.342 |
| 4.546272965 | 36715.740 |
| 4.496377744 | 30598.355 |
| 4.486100396 | 35178.147 |
| 4.511561155 | 31360.122 |
| 4.548141639 | 30626.714 |
| 4.564852289 | 32475.897 |
| 4.485698078 | 35329.837 |